Mechatronic Actuators Trainer

Fundamental Labs – Brushed DC Motor

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QUANSER
INNOVATE·EDUCATE

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Caution

This equipment is designed to be used for educational and research purposes and is not intended for use by the public. The user is responsible for ensuring that the equipment will be used by technically qualified personnel only. Users are responsible for certifying any modifications or additions they make to the default configuration.

FCC Notice This device complies with Part 15 of the FCC rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.
Note: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment.

Industry Canada Notice This Class A digital apparatus complies with CAN ICES-3 (A). Cet appareil numérique de la classe A est conforme à la norme NMB-3 (A) du Canada.
Japan VCCI Notice This is a Class A product based on the standard of the Voluntary Control Council for Interference (VCCI). If this equipment is used in a domestic environment, radio interference may occur, in which case the user may be required to take corrective actions.

この装置は、クラス A 情報技術装置です。この装置を家庭環境で使用すると電波妨害を引き起こすことがあります。この場合には使用者が適切な対策を講ずるよう要求されることがあります。 VCCI–A

Waste Electrical and Electronic Equipment (WEEE)
This symbol indicates that waste products must be disposed of separately from municipal household waste, according to Directive 2012/19/EU of the European Parliament and the Council on waste electrical and electronic equipment (WEEE). All products at the end of their life cycle must be sent to a WEEE collection and recycling center. Proper WEEE disposal reduces the environmental impact and the risk to human health due to potentially hazardous substances used in such equipment. Your cooperation in proper WEEE disposal will contribute to the effective usage of natural resources.

电子信息产品污染控制管理办法 (中国 RoHS)

中国客户Quanser Consulting Inc. 关于关于限制在电子电气设备中使用某些有害成分的指令 (RoHS)。

CE Compliance

This product meets the essential requirements of applicable European Directives as follows:
- 2014/30/EU; Electromagnetic Compatibility Directive (EMC)

Warning: This is a Class A product. In a domestic environment this product may cause radio interference, in which case the user may be required to take adequate measures.

Brushed DC Motor Lab

Why Explore a Brushed DC Motor?

There are two types of commonly used DC motors: brushless and brushed. This lab will help you explore brushed DC motors. Brushed DC motors are less efficient, and less durable compared to Brushless DC motors due to power losses and the wear and tear of brushes and the commutator. These brushed motors use a rotating coil that is magnetized using DC current, interacting with stationary magnets around the central coil. This simple design makes Brushed DC motors cost-effective and easy to manufacture and drive. They are commonly used in applications that require a high starting torque.

Brushed DC motors are often used in applications that do not require constant operation, such as power tools, household appliances and automative systems (power windows, seat adjusters, windshield wipers, etc.). As such, prolonged operation resulting in overheating is not a major concern.

Background

This lab is part of the Fundamentals labs for the Mechatronic Actuators Trainer. These labs will guide you through four different types of motors, so you can understand their operating principle and get you comfortable with using them. This will also include highlighting datasheets to learn how to interpret them and know what to look for.

Prior to starting this lab, please review the following concept review (located in Documents/Quanser/4_concept_reviews/),

- 4_concept_reviews/Mechatronics/**002_motor_dc**.
- 4_concept_reviews/Mechatronics/**007_rotary sensors** **(Rotary Encoders section)**.

Getting Started

In this lab, you will use the **Actuators Trainer** to learn about the basics of **brushed DC motors**. The lab will guide you through the basic model and characteristics of such a motor and compare motor performance to specifications from the datasheet.

Before you begin this lab, ensure that the following criteria are met:

- Make sure you have either the provided Brushed DC motor, or another motor properly wired to be able to interface with the Actuators Trainer.
- The provided DC motor is this one. Confirm your motor is wired properly to the 2-pin connector to run it and the 5-pin encoder connector with an empty space.

- Make sure the Actuators Trainer has been setup and tested. See the Quick Start Guide (Quanser/2_quick_start_guides/actuators_trainer) for details on this step.
- You are familiar with the basics of the programming language you are using for this lab.
- Have your development environment ready,
    - If using Python, we recommend using Visual Studio Code.